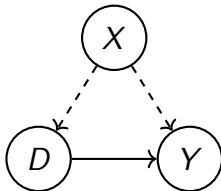


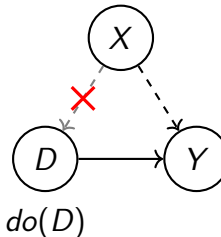
- 1 Opening remarks
- 2 Selected topics in experimental research
- 3 Conclusion

The core problem of causal inference



Confounding path: $D \leftarrow X \rightarrow Y$

Randomized / manipulated version



Randomization breaks the arrow into D

- 1 Opening remarks
- 2 Selected topics in experimental research
- 3 Conclusion

Common practices in experimental research

Think about what you usually do in designing and analyzing experiments.

Goal of this session

Experiments are powerful

The idea of experimentation is great: we break confounding paths with randomization and obtain more credible causal inference.

...But the devil is in the implementation: we often use additional practices to resolve some issues.

Goal of this session

Experiments are powerful

The idea of experimentation is great: we break confounding paths with randomization and obtain more credible causal inference.

...But the devil is in the implementation: we often use additional practices to resolve some issues.

The deeper questions

- Why do these practices increase validity?
- What assumptions do they add?
- When might they change the causal question?

Goal of this session

Experiments are powerful

The idea of experimentation is great: we break confounding paths with randomization and obtain more credible causal inference.

...But the devil is in the implementation: we often use additional practices to resolve some issues.

Goal: use the new language of cause and effect – DAGs – to diagnose, understand, and improve experimental practices.

Disclaimer

1. These practices are not necessarily bad.

The point is not that current practices are wrong, but to understand 1) **why we use them**, 2) **when they help**, and 3) **when they may create new problems**.

2. Publication norms still matter

For publication, follow the commonly accepted practices of the journals you target.

This session is to deepen your understanding and prepare you for future methodological updates.

Structure of the discussion

Common Practice → Ideal DAG → Altered DAG → Simulation → Best Practice

For each experimental practice, we will ask:

- 1 What do researchers commonly do?
- 2 What is the ideal causal story?
- 3 How does the practice change the DAG?
- 4 What are the consequences? *Can we simulate them?*
- 5 What should we do instead, or do more carefully?

Goal: move from ritualized practice to causal diagnosis.

① Opening remarks

2 Selected topics in experimental research

Randomization

Treatment vs. construct effect

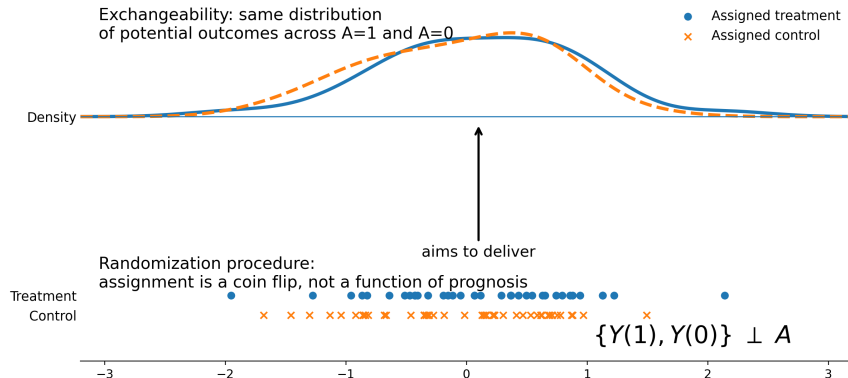
Data deletion after checks

3 Conclusion

- 1 Opening remarks
- 2 Selected topics in experimental research
 - Randomization
 - Treatment vs. construct effect
 - Data deletion after checks
- 3 Conclusion

Key distinction

Randomization is the **procedure**;
Exchangeability is the **property** needed for causal inference.

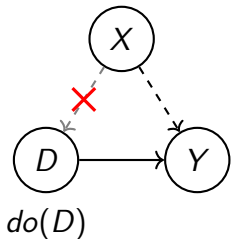


Altered DAG: realized imbalance

Problem

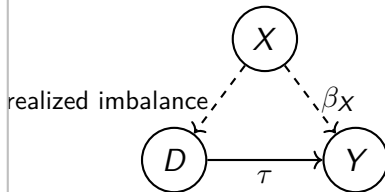
Even with a valid randomization procedure, a finite sample can produce **realized imbalance**.

Ideal randomization



Randomization breaks the arrow into D

Realized DAG



Exchangeability is the property we need

Altered DAG: realized imbalance

Problem

Even with a valid randomization procedure, a finite sample can produce **realized imbalance**.

DAAG diagnosis

The randomization procedure was valid, but in the realized sample:

$$r_{D,X} \neq 0$$

If X also affects Y , the treatment estimate can contain omitted-variable bias:

$$\hat{\tau} \approx \tau + \beta_X r_{D,X}$$

Simulation: randomization vs. realized exchangeability

- 1 Draw baseline covariate: $X_i \sim \mathcal{N}(0, 1)$.
- 2 Randomize treatment:
 $D_i \sim \text{Bernoulli}(0.5)$.
- 3 Generate outcome:

$$Y_i = \tau D_i + \beta_X X_i + \varepsilon_i$$

- ④ Estimate:

$$Y_j \sim D_j \quad \text{and} \quad Y_j \sim D_j + X_j$$

What varies across simulations?

Even though D is randomized, in a small sample: $r_{DX} = \text{cor}(D, X)$ can be nonzero by chance.

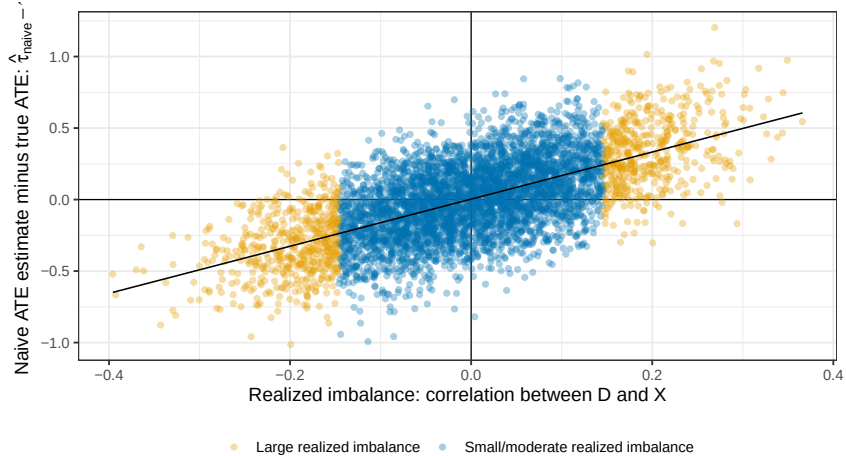
Main idea

When X predicts Y , realized imbalance in X can distort the naive estimate:

$$\hat{\tau}_{\text{naive}} - \tau \approx \beta_X(\bar{X}_1 - \bar{X}_0)$$

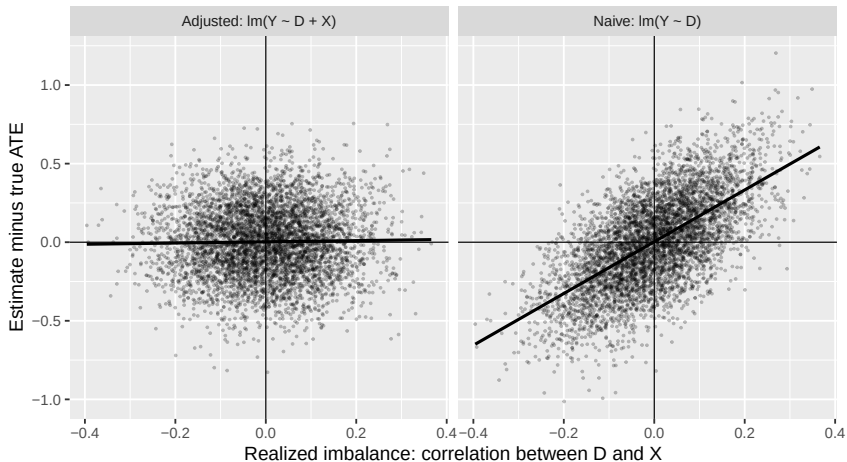
Simulation: randomization and realized imbalance

Please find the codes here: [CLICK HERE](#).



Simulation: randomization and realized imbalance

Please find the codes here: [CLICK HERE](#).



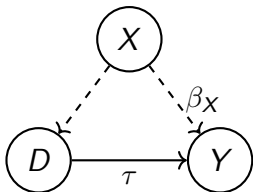
Larger sample size can help, but the imbalance still exists.



Balance checks and regression adjustment

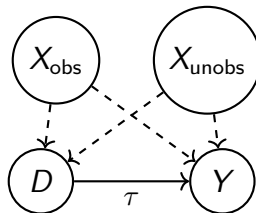
You can only balance / adjust on variables you measured.
Unmeasured causes of Y remain invisible.

Realized DAG



Exchangeability is the property we need

Observed and unobserved X



Some sources of imbalance may be unobserved

- 1 Opening remarks
- 2 Selected topics in experimental research
 - Randomization
 - Treatment vs. construct effect
 - Data deletion after checks
- 3 Conclusion

Some designs ¹ for the manipulation feel “clever” (CLICK FOR MORE EXAMPLES).

What makes these smart?

We get **treatment effect** from experiments (treatment D on Y).

$$D \rightarrow Y$$

What makes these smart?

We get **treatment effect** from experiments (treatment D on Y).

$$D \rightarrow Y$$

We want to test psych theory or **construct effect** (psych construct M on Y).

$$D \rightarrow M \rightarrow Y$$

What makes these smart?

We get **treatment effect** from experiments (treatment D on Y).

$$D \rightarrow Y$$

We want to test psych theory or **construct effect** (psych construct M on Y).

$$D \rightarrow M \rightarrow Y$$

Question

Are we estimating the effect of the manipulation, or the effect of the psychological construct?

Common practice: manipulation as construct

We say that:

"We manipulated anxiety."

"We manipulated trust."

"We manipulated perceived scarcity."

"We manipulated identity threat."

But more precisely:

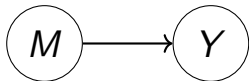
We manipulated a **stimulus** intended to change anxiety / trust / scarcity / identity threat.

Common practice: $D = M$

Causal reality: $D \rightarrow M$

The ideal theoretical DAG

Ideal theoretical object



$$M \rightarrow Y$$

$D = \text{manipulation}$

M = latent psychological construct

$$Y = \text{outcome}$$

Reality in experiments



$$D \rightarrow M \rightarrow Y$$

Randomization gives us leverage on $D \rightarrow Y$,
but the theory usually lives in $M \rightarrow Y$.

Ideal DAG



```

graph LR
    D((D)) --> A((A))
    D --> B((B))
    D --> M((M))
    A --> Y((Y))
    B --> Y
    M --> Y
    style A stroke-dasharray: 5 5
    style B stroke-dasharray: 5 5
    style M stroke-dasharray: 5 5
    style D fill:#fff,stroke:#000,stroke-width:1px
    style Y fill:#fff,stroke:#000,stroke-width:1px
  
```

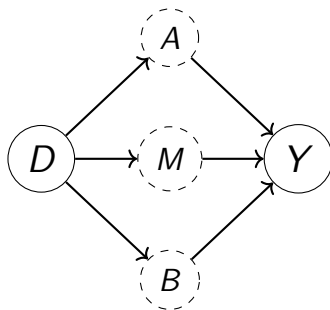
The manipulation may move many things.

Ideal manipulation: the “surgical intervention” fantasy

Examples:

- Fear induction: attention, arousal, demand awareness, memory, mood.
- Eco-label: sustainability, quality, naturalness, social approval, price expectations.
- Scarcity cue: urgency, popularity, reactance, suspicion.

Altered DAG in practice

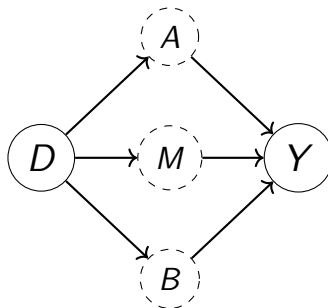


The manipulation may move many things.

Ideal manipulation: the “surgical intervention” fantasy

The randomized treatment effect is still causal, but it is causal for the whole bundle $(A, M, B \dots)$, not necessarily for (M) .

Altered DAG in practice



The manipulation may move many things.

The design dilemma: strong versus subtle manipulations

Manipulation type	Advantage	Risk
Strong manipulation	Large first stage: $D \rightarrow M$	Also changes M' , demand, suspicion, attention
Subtle manipulation	Less demand and fewer obvious side pathways	May not move M enough
Clean manipulation	Best for causal interpretation	Often hard to design

**Psychological manipulations face a
relevance–exclusion tradeoff.**

Relevance: $D \rightarrow M$ vs. **Exclusion:** $D \rightarrow Y$ only through M

Manipulation checks: helpful, but not enough

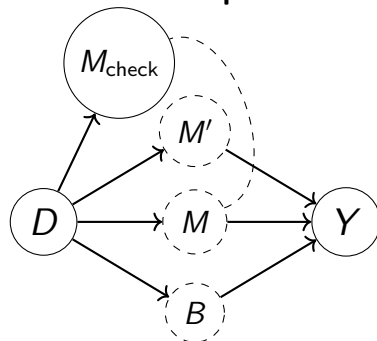
Common practice:

Show that $D \rightarrow M_{\text{check}}$

But it does **not** prove: $M \rightarrow Y$.

And it does **not** rule out: $D \rightarrow M' \rightarrow Y$.

The bundle problem



A successful manipulation check establishes relevance, not exclusion.

Best practices: manipulation checks

1. Check relevance

$$D \rightarrow M_{\text{check}}$$

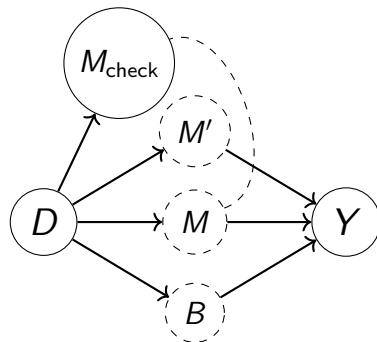
Does the manipulation change the measured construct it was intended to move?

2. Check exclusion

$$D \nrightarrow M' \rightarrow Y$$

Does the manipulation also influence the outcome through other psychological channels?

The bundle problem



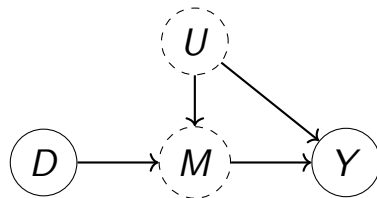
Confounding of the construct M

Even if D is randomized, M itself not.

Examples of U :

- baseline personality
- prior attitudes
- mood
- cognitive ability
- political ideology
- social desirability

Construct-level confounding^a



Rand D does not randomize M .

So the (measured) M and Y may still be confounded.

^aThis connects naturally to mediation analysis, which we will discuss in more detail in **Lecture 3**.

① Opening remarks

② Selected topics in experimental research

Randomization

Treatment vs. construct effect

Data deletion after checks

③ Conclusion

Data deletion after checks

Quick question

Have you ever done one of these before analyzing an experiment?

- Dropped failed attention checks
- Dropped failed manipulation checks
- Dropped failed comprehension checks
- Dropped speeders
- Dropped straightliners
- Dropped outliers
- Dropped suspicious respondents
- Re-ran analysis after exclusions

A common practice

This is common practice:

The screenshot shows the Journal of Consumer Research (JCR) website. The header includes the JCR logo and navigation links: Issues, More Content, Publish, Purchase, Alerts, About, and a search bar. The search results for 'outlier' show 1-20 of 180 results. The first result is titled 'outlier' and is from the article 'COVARIATE-ADJUSTED AMOUNT OF FOOD CONSUMED, EXPERIMENT 3 NOTE.—Outliers we...'. The article is published on 28 May 2018. A small bar chart is visible on the right side of the article preview, showing the distribution of food consumed across different conditions. The chart has a y-axis labeled 'Average food consumed (g)' and an x-axis labeled 'Outlier level'. It compares 'Control condition' (white bars) and 'Outlier condition' (black bars). The 'Outlier condition' shows a significantly higher average food consumed compared to the 'Control condition'.

J C R
Journal of Consumer Research

Issues More Content ▾ Publish ▾ Purchase Alerts About ▾ Journal of Consumer Research Search

Modify your search

outlier

☐ Full Text ☐ Keywords
☐ Title ☐ Author/Editor/ORCID
☐ Abstract ☐ References/Bibliography

Update + Add term

Availability

☐ All content I have access to
☐ Purchased
☐ Open Access

1-20 of 180

outlier

[Save search](#) [Export title list \(.csv\)](#)

Sort by Relevance ▾ 1 2 3 4 5 Next

COVARIATE-ADJUSTED AMOUNT OF FOOD CONSUMED, EXPERIMENT 3 NOTE.—Outliers we...

[in Guilt Dynamics: Consequences of Temporally Separating Decisions and Actions](#)

Published: 28 May 2018

FIGURE 7 COVARIATE-ADJUSTED AMOUNT OF FOOD CONSUMED, EXPERIMENT 3 NOTE.—**Outliers** were excluded based on the median absolute deviation method (see

Image

P

Figure 1: Search results of outliers deletion

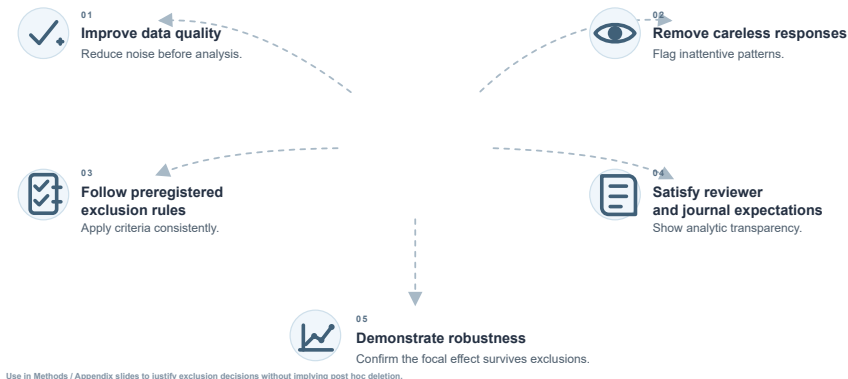
A common practice

This is common practice:

Among our sample (of 236 responses)², 88% reported always or often using attention checks in their work as a means of excluding low quality responses... (Perfecto and O'Donnell, 2025).

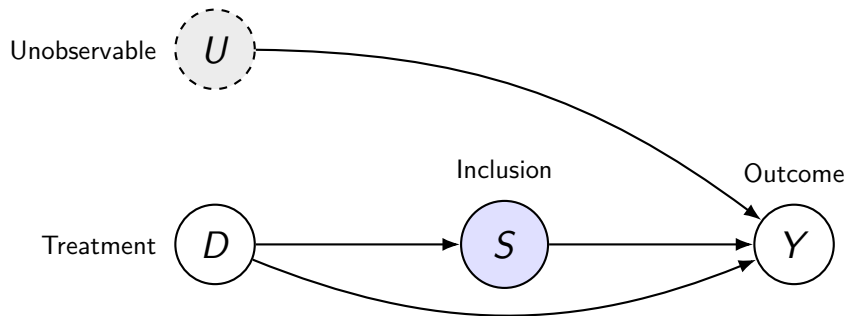
²The majority of respondents reported affiliations with marketing departments (62%), followed by psychology departments (22%) and other business school departments (12%), with a small group of researchers in industry (4%)

The key question to data deletion



The key question is not: “Are these practices bad?”
...but rather: **what causal estimand do they create?**

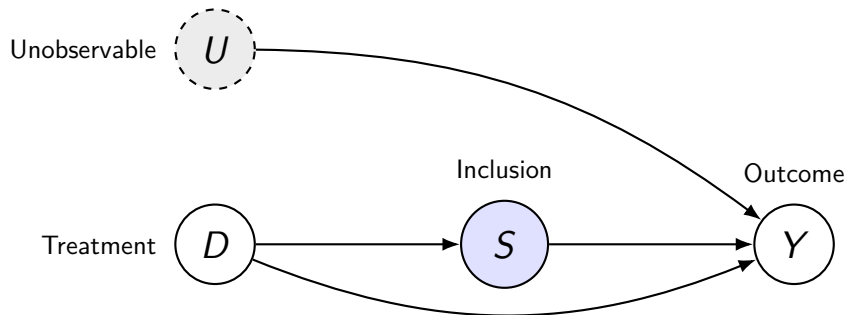
The ideal DAG



Key idea

When we delete data after a check, we are not just “cleaning” the dataset. We are conditioning on being included in the final analysis.

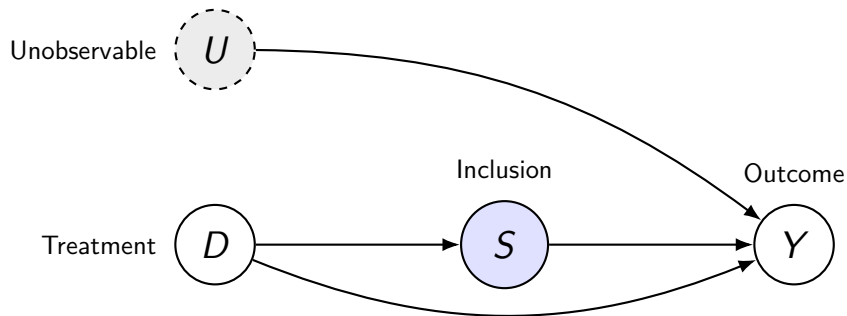
The ideal DAG



Ideal DAG

Exclusion happens after treatment, but selection is not driven by the unobservable – no direct $U \rightarrow S$ path.

The ideal DAG



Inclusion decision S

By the DAG, excluding data means “conditional on S .” This blocks the $D \rightarrow S \rightarrow Y$ path, and we obtain $P(Y \mid \text{do}(D), S = 1)$.

Outliers: deletion rule = selection variable S

Key idea

Every deletion rule creates a selection variable.

Outlier deletion: different rules, different S

- Drop observations beyond ± 3 SD
- Drop top/bottom 1% or 5%
- Drop based on Cook's distance or leverage
- Drop speeders based on completion time
- Drop “implausible” or “inconsistent” responses

DAG question: what causes someone to be labeled an outlier?

Attention checks = selection variable S

Straightline

For each item, indicate the extent you have felt this way over the past week.

	Not at all	A little	Moderately	Quite a bit	Extremely
Inspired	☐	☐	☐	☐	☐
Active	☐	☐	☐	☐	☐
Excited	☐	☐	☐	☐	☐
Hostile	☐	☐	☐	☐	☐
Ashamed	☐	☐	☐	☐	☐
Irritable	☐	☐	☐	☐	☐
Interested	☐	☐	☐	☐	☐
Proud	☐	☐	☐	☐	☐
Select "not at all"	☐	☐	☐	☐	☐

English Comprehension

The trophy doesn't fit into the brown suitcase because it's too small. What is too small?

Self-Report

How closely did you pay attention to the instructions and tasks during the experiment?

1 not at all	2	3	4	5 I gave the tasks my undivided attention
-----------------	---	---	---	---

Instructional Manipulation Check (IMC)

Imagine bringing home a new puppy to your family. What would you name it? Data from decades of pet records show that pet names come in and out of style, just like children's names. Like children's names, pet name trends are often influenced by popular culture. These data are less useful though, when pet owners don't pay attention to the surveys. To show you are paying attention to this one, please click Other and write: reading is good.

Fido	Gus
Chewy	Pal
Barbers	Clayton
Anderson	Other
St Woolfington	

Obvious

How often do you breathe oxygen?

1 never ☐	2 a few times a year ☐	3 once a month ☐	4 once a week ☐	5 always ☐
-------------------------------------	--	---	--	--------------------------------------

Memory

In the survey you just completed you read about a student who completed an essay for class. What was the essay about?

Opposing capital punishment
Supporting capital punishment
Opposing capital gains tax
Supporting capital gains tax
The proposed capital of the Midwest
I don't remember

Figure 2: Commonly Used Attention Checks (Perfecto and O'Donnell, 2025)

Different checks, different psychological processes

Memory check

Memory ability $\rightarrow S$

May capture:

- attention
- encoding
- recall ability
- task engagement

English comprehension

Comprehension $\rightarrow S$

May capture:

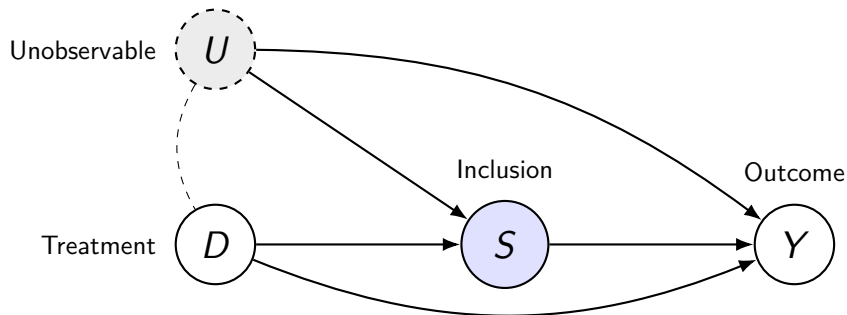
- attention
- language ability
- careful reading
- compliance

DAG diagnosis

If these processes also affect Y , then we have a problem.

The DAG move: deletion means selection

Is S caused by U or not?

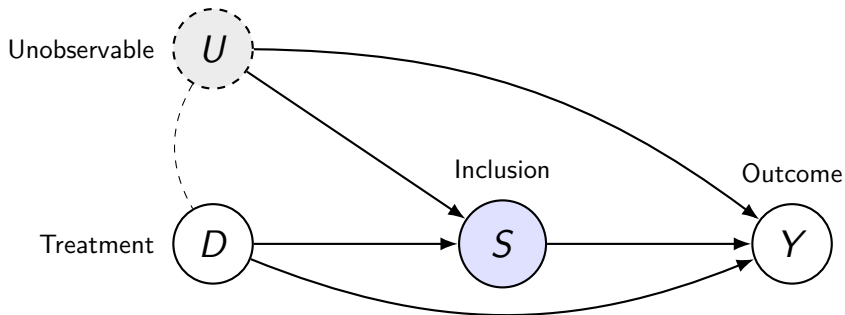


If yes, deletion may change the experiment – **the collider problem**.

The DAG move: deletion means selection

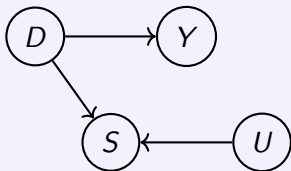
Two key questions:

- Is selection S caused by some unobserved variable U ?
- Does that same U also influence the outcome Y ?



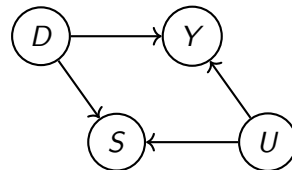
Example 1: outlier deletion

Case A: Less problematic



e.g., delete participants who enter impossible WTP, such as -\$999 or \$999999.

Case B: Problematic

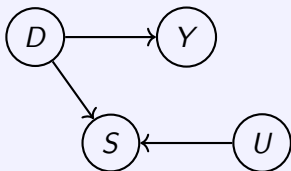


e.g., delete participants whose willing-to-pay (WTP) is more than 3 SD above the mean.

DAG question: what makes someone an outlier, and does that also influence Y?

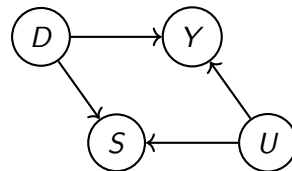
Example 2: Attention-Check Deletion

Case A: Less problematic



Example: a purely random browser glitch or accidental click.

Case B: Problematic



Examples of U :

- memory ability
- need for cognition
- involvement

DAG question: what makes someone a nonattender, and does that also influence Y ?

Consistency before vs. after deletion

Researchers often compare results before and after deletion.

- Similar results $\Rightarrow$ deletion is harmless.
- Different results $\Rightarrow$ deletion is problematic

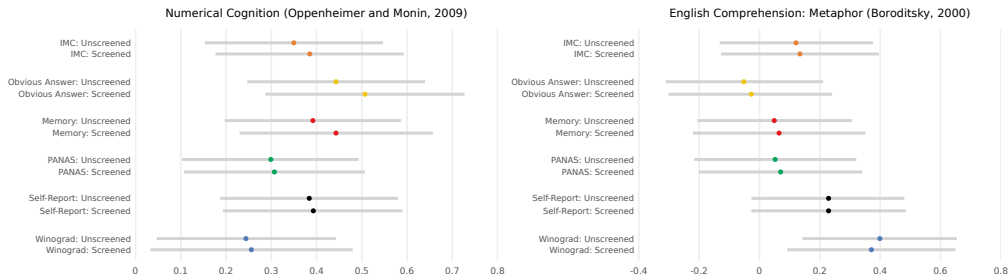


Figure 3: Consistency of treatment effect; reproduced from Perfecto and O'Donnell (2025)

Check the consistency interpretation with simulations

Scenario A: random deletion

$$D \rightarrow Y$$

$$S \perp D, U, Y$$

Selection is as-if random. Identification remains valid.

But after deletion, the estimate may still change because of sampling variation.

Scenario B: collider deletion

$$D \rightarrow S \leftarrow U \rightarrow Y \quad D \rightarrow Y$$

Selection depends on treatment and an unobserved trait that also affects the outcome.

Identification is not valid after deletion.

But after deletion, the estimate may still show consistency.

Simulation procedure

For each simulated experiment

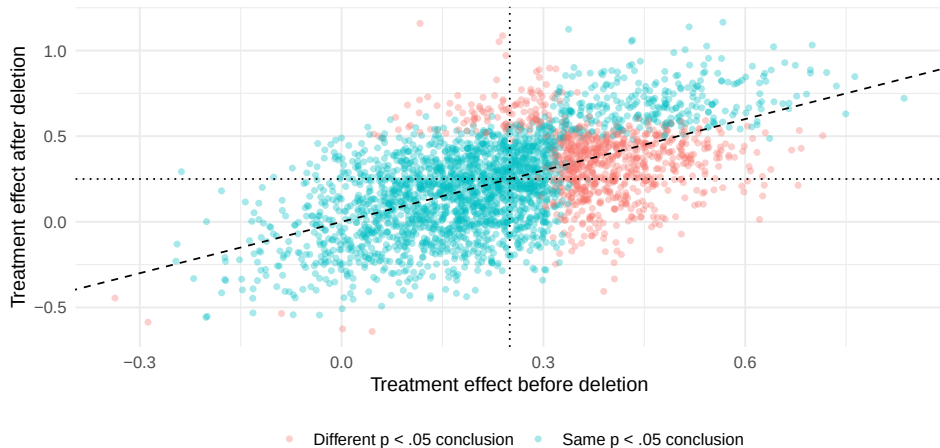
- ① Estimate treatment effect in the full sample: $\hat{\tau}_{full}$
- ② Delete observations using S
- ③ Estimate treatment effect in the retained sample: $\hat{\tau}_{kept}$
- ④ Compare whether the two estimates look “robust”

The codes can be downloaded from: [CLICK HERE](#).

Results: random deletion

A. Random deletion: identification is fine

But estimates and p-value conclusions can still change after deletion



B. Collider deletion: identification is not fine
Yet before/after estimates often look statistically consistent



Best practices: data deletion after checks

1. Deletion can help, but it can also change identification

Deletion may remove noise and improve data quality.

But deletion is also selection:

$S = 1$ if kept in the analysis

The key question is:

What causes S ?

and

Do those causes also affect Y ?

Best practices: data deletion after checks

2. Consistency is not definitive proof

Consistency before and after deletion is useful, but not definitive. Better interpret consistency as plausibility and supporting evidence.

$$\hat{\tau}_{full} \approx \hat{\tau}_{kept} \not\Rightarrow \text{deletion is harmless}$$

$$\hat{\tau}_{full} \neq \hat{\tau}_{kept} \not\Rightarrow \text{deletion is biased}$$

Best practices: data deletion after checks

3. Best-case solution

When possible, retain the “problematic” observations and use bounds / set identification:

Best-case scenario $\leq \tau \leq$ Worst-case scenario

More on this in Lecture 5.

- 1 Opening remarks
- 2 Selected topics in experimental research
- 3 Conclusion

High-level takeaways

DAGs as a design language

DAGs help us deepen our understanding of experimental research, quasi-experimental designs, and other research designs.

Do not take common practices for granted

Randomization, manipulation checks, attention checks, deletion rules, covariate adjustment, and mediation analysis are not just procedures.

They encode assumptions about cause and effect.

High-level takeaways

Always ask why

Why does this practice help?

What assumption does it require?

What causal structure does it create?

The moment you start drawing a DAG, you are already thinking more deeply about what works, what does not, and why.

